There are 5 main methods to detect multicollinearity in a dataset. Multicollinearity occurs when two or more independent variables (features) are highly correlated with each other.

1. Correlation Matrix (Heatmap)

A correlation matrix helps identify strong relationships between pairs of independent variables.

Code:

```
import seaborn as sns

import matplotlib.pyplot as plt

X = dataset[["ssc_p", "hsc_p", "degree_p", "etest_p", "mba_p"]]

sns.heatmap(X.corr(), annot=True, cmap="coolwarm")

plt.show()
```

Interpretation:

- Correlation close to +1: Strong positive relationship.
- Correlation close to -1: Strong negative relationship.
- Correlation close to 0: Weak linear relationship.
- Absolute correlation above 0.8: Potential multicollinearity that should be investigated.

2. VIF (Variance Inflation Factor)

VIF measures how strongly one independent variable can be explained by the other independent variables. It is one of the most commonly used methods for detecting multicollinearity.

Code:

```
import pandas as pd

from statsmodels.stats.outliers_influence import variance_inflation_factor

X = dataset[["ssc_p", "hsc_p", "degree_p", "etest_p", "mba_p"]]

vif = pd.DataFrame()

vif["Variable"] = X.columns

vif["VIF"] = [ variance_inflation_factor(X.values, i)for i in range(X.shape[1])]

print(vif)
```

Interpretation:

VIF value	Meaning
1	No multicollinearity with the other variables
1–5	Usually acceptable
5–10	Potential multicollinearity
Above 10	High multicollinearity

3. Tolerance

Tolerance is the reciprocal of VIF. It helps identify how much of a variable's variance is not explained by the other independent variables.

$$\text{Tolerance} = \frac{1}{\text{VIF}}$$

Interpretation:

- Tolerance below 0.2: Investigate potential multicollinearity.
- Tolerance below 0.1: May indicate serious multicollinearity.

You can calculate it using:

Code:

```
vif["Tolerance"] = 1 / vif["VIF"]

print(vif)
```

4. Condition Number

The condition number helps assess multicollinearity and numerical instability in the feature matrix.

Code:

```
import numpy as np

X = dataset[["ssc_p", "hsc_p", "degree_p", "etest_p", "mba_p"]]

print(np.linalg.cond(X))
```

A high condition number can indicate multicollinearity, but it is affected by the scale of the variables. Standardizing the features before calculating it makes the result easier to interpret.

5. Regression Coefficient Analysis

Multicollinearity can also affect regression model results. Possible warning signs include:

- Regression coefficients changing significantly when features are added or removed.
- Unexpected coefficient signs.
- Large standard errors for regression coefficients.
- Individual predictors appearing statistically insignificant despite the model having strong overall explanatory power.

These are warning signs rather than definitive proof of multicollinearity.

Which method should you use?

Method	**Main purpose**
Correlation matrix	Detect strong relationships between pairs of features
VIF	Detect multicollinearity involving one feature and the other features
Tolerance	Alternative way to interpret VIF
Condition number	Assess overall multicollinearity and numerical instability
Regression coefficient analysis	Understand how multicollinearity may affect model estimates